

CO5-AT-3

MLA0201-Fundamentals Of Machine Learning

Mock Interview

Q1. How would you use Bayesian Networks to predict machine failures?

Answer)

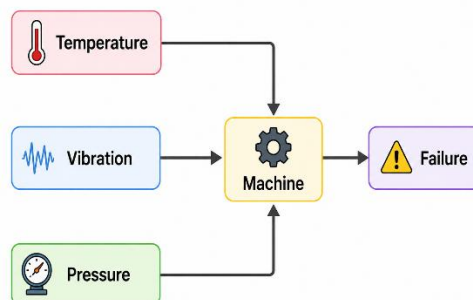
A **Bayesian Network (BN)** is a directed graphical model that represents relationships between variables using nodes and directed edges.

For an industrial monitoring system, I would create nodes such as:

- Temperature
- Vibration
- Pressure
- Lubrication
- Machine Age
- Machine Failure

The network would represent dependencies between these variables. Historical machine data would be used to learn the conditional probability tables.

For example:



When new sensor readings are received, Bayesian inference calculates:

$$P(F, a, ilure \mid TemperatureVibrationPressure)$$

If the calculated probability exceeds a predefined threshold, the system can generate an early warning.

Example:

If high temperature and high vibration are detected, the network may calculate a high probability of machine failure.

Conclusion:

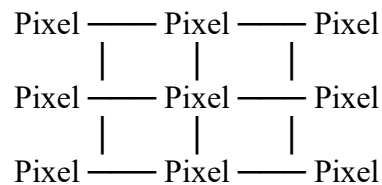
Bayesian Networks help predict machine failures by combining sensor evidence and probabilistic relationships, allowing early fault detection and preventive maintenance.

Q2. How would Markov Random Fields help model neighborhood pixel dependencies in image denoising?

Answer)

A Markov Random Field (MRF) is an undirected graphical model used to represent relationships between neighboring variables.

In image denoising, each pixel is represented as a node. Neighboring pixels are connected because nearby pixels usually have similar properties.



The observed noisy image is used as evidence, while the MRF estimates the most likely clean pixel values.

The model considers:

1. The observed noisy pixel.
2. The values of neighboring pixels.
3. Smoothness between neighboring pixels.
4. The probability of a clean image.

The objective can be represented as:

$$E(X) = \sum_i D_i(X_i) + \sum_{i,j} V_{ij}(X_i, X_j)$$

where D_i represents the data cost and V_{ij} represents the neighborhood relationship.

Example:

If one pixel is very different from all its surrounding pixels, the MRF may classify it as noise and replace it with a value consistent with its neighbors.

Conclusion:

MRFs are useful for image denoising because they explicitly model spatial dependencies between neighboring pixels, producing smoother and cleaner images while preserving important structures.

Q3. How would you apply a Hidden Markov Model to process sequential audio signals in speech recognition?

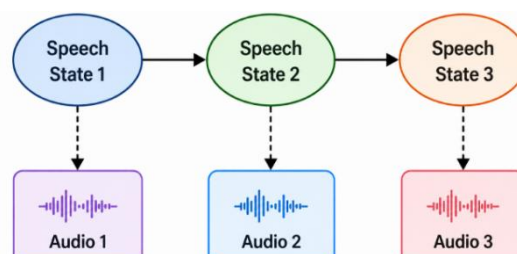
Answer)

A **Hidden Markov Model (HMM)** is a probabilistic model used for sequential data where the actual states are hidden but generate observable outputs.

In speech recognition:

- **Hidden states** → phonemes or speech units
- **Observations** → audio features
- **Transitions** → movement between speech states
- **Emission probabilities** → probability of an audio feature being produced by a state

A simple structure is:



The process is:

1. Capture the speech signal.
2. Divide it into small time frames.
3. Extract audio features.
4. Feed the features into the HMM.
5. Calculate the most likely sequence of hidden states.
6. Convert the states into recognized words.

The **Viterbi algorithm** can be used to find the most likely state sequence.

Example:

For spoken word "CAT", the HMM processes successive audio frames and identifies the most probable phoneme sequence.

Conclusion:

HMMs are suitable for speech recognition because they model the temporal sequence and uncertainty present in audio signals.

Q4. How would you explain the use of Conditional Random Fields for POS tagging?

Answer

A Conditional Random Field (CRF) is an undirected probabilistic model used for sequence labeling. It predicts labels based on the complete input sequence.

For Part-of-Speech (POS) tagging, each word is an observation and its POS tag is the output label.

Example:

Words: The student reads
Tags: DET NOUN VERB

A CRF considers:

- The current word
- Neighboring words
- Previous and next labels
- Word features
- Capitalization
- Prefixes and suffixes

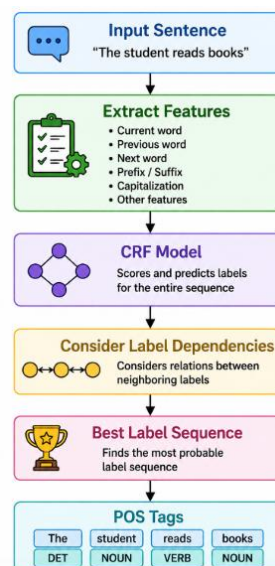
The CRF estimates:

$$P(Y | X)$$

where:

- X = observed word sequence
- Y = POS-tag sequence

Working:



Example:

The word "**book**" can be a noun or verb. CRF can use surrounding words to determine the correct tag.

Conclusion:

CRFs are effective for POS tagging because they consider relationships between neighboring labels and features of the complete input sequence.

Q5. How does conditional independence simplify probabilistic inference?**Answer)**

Conditional independence means that two variables become independent when another variable is known.

For example:

Rain → Wet Road

Rain → Traffic

Here, Wet Road and Traffic can be conditionally independent given Rain:

$$P(WetRoadTraffic | Rain) = P(WetRoad | Rain)P(Traffic | Rain)$$

Without conditional independence, the model would need to represent a much larger joint probability distribution.

Why does it simplify inference?

1. Reduces the number of probabilities that must be calculated.
2. Reduces memory requirements.
3. Makes probability calculations faster.
4. Allows complex models to be divided into smaller parts.
5. Makes graphical models easier to understand.

Example:

If:

Disease → Fever

Disease → Cough

Disease → Fatigue

then, given the disease, the symptoms can be modeled using separate conditional probabilities.

Conclusion:

Conditional independence simplifies inference by allowing a large probability distribution to be factorized into smaller and easier-to-compute components.

Q6. Compare directed and undirected graphical models for representing symptom relationships.**Answer)**

Directed graphical models and undirected graphical models represent relationships between variables differently.

Feature	Directed Graphical Model	Undirected Graphical Model
Structure	Directed edges	Undirected edges
Example	Bayesian Network	Markov Random Field
Represents	Directional dependencies	Mutual relationships
Probability	Conditional probability	Potential functions
Common use	Diagnosis and causal reasoning	Spatial/neighbor relationships

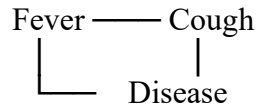
Directed Example:

Disease



Fever Cough

This can represent a dependency from disease to symptoms.

Undirected Example:

It represents relationships without specifying a direction.

Healthcare Application:

A Bayesian Network is useful when disease-to-symptom dependencies or causal relationships need to be represented.

An MRF is useful when symptoms have strong mutual relationships and direction is not important.

Conclusion:

Directed models are generally better for representing directional or causal dependencies, while undirected models are useful for representing symmetric relationships and dependencies.

Q7. How would you perform Bayesian inference when patient symptom data is partially missing?**Answer)**

When some patient symptoms are missing, Bayesian inference can still estimate the probability of each possible diagnosis using the available evidence.

Suppose:

Available:

Fever = Yes

Cough = Yes

Missing:

Fatigue

The system does not need to assign an arbitrary value to the missing symptom. Instead, it marginalizes over the missing variable.

The diagnosis can be calculated as:

$$P(\text{Disease} \mid \text{Evidence}) \propto P(\text{Disease}) \times P(\text{Evidence} \mid \text{Disease})$$

If a missing variable X exists:

$$P(D \mid E) = \sum_X P(D, X \mid E)$$

Steps:

1. Collect available symptoms.
2. Identify missing symptoms.
3. Use the Bayesian Network.
4. Marginalize over missing variables.
5. Calculate posterior probability for each disease.
6. Select the diagnosis with the highest probability.

Example:

If the available evidence strongly supports flu, the system can still estimate:

Flu → 0.75

Cold → 0.18

Allergy → 0.07

even though one symptom is missing.

Conclusion:

Bayesian inference handles missing data naturally by using available evidence and marginalizing over unknown variables, allowing diagnosis without requiring every symptom to be present.

Q8. How would you use an HMM to analyze customer purchase sequences?**Answer)**

An HMM can model customer behavior over time by treating the customer's actual shopping state as hidden and purchases as observations.

Hidden states:

Browsing → Interested → Buying → Loyal

Observations

View Product

Add to Cart

Purchase

Repeat Purchase

The model learns:

- Initial state probabilities
- State transition probabilities
- Observation/emission probabilities

Working:

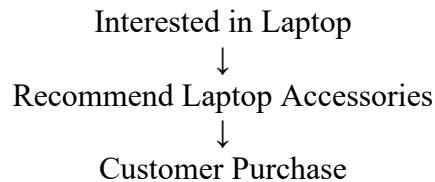
For example, a customer who repeatedly views laptops and adds one to the cart may have a high probability of being in an Interested state.

The system can predict the next likely purchase behavior.

Recommendation Application:

The recommendation system can use the predicted state to recommend suitable products.

For example:



Conclusion:

HMMs can model sequential customer behavior and help e-commerce systems predict future purchases and provide personalized recommendations.

Q9. Why do CRFs often outperform HMMs in Named Entity Recognition?

Answer)

Both HMMs and CRFs can be used for sequence labeling, but CRFs often perform better in Named Entity Recognition (NER).

HMM

HMM models:

$$P(X, Y)$$

and generally makes strong assumptions about how observations and states are generated.

CRF

CRF directly models:

$$P(Y | X)$$

This allows CRFs to use many overlapping features of the input sentence.

Example:

"Dr. Ravi works at Microsoft."

CRF can consider:

- Word itself
- Previous word
- Next word
- Capitalization
- Prefix
- Suffix
- Position
- Neighboring labels

For example:

Dr. → TITLE

Ravi → PERSON

Microsoft → ORGANIZATION

Why CRF can perform better:

1. Uses multiple features simultaneously.
2. Does not require strong independence assumptions about observations.
3. Models dependencies between neighboring labels.
4. Handles overlapping and contextual features effectively.
5. Produces a globally optimized label sequence.

Conclusion:

CRFs often outperform HMMs in NER because they can use rich contextual features and label dependencies while directly modeling the conditional probability of the label sequence.

Q10. How would graphical model learning and generalization enhance fraud detection?**Answer)**

Graphical models can represent relationships between variables involved in financial transactions and learn these relationships from historical data.

Possible variables include:

Transaction Amount



Location → Fraud



Time



Device

The model can learn patterns from both legitimate and fraudulent transactions.

Learning:

Historical transaction data is used to estimate:

- Probabilities
- Dependencies
- Model parameters
- Fraud patterns

For a new transaction, the model calculates:

$$P(\text{Fraud} \mid \text{Transaction Evidence})$$

Generalization:

Generalization means that the learned model should perform well on new, unseen transactions, not only on training data.

To improve generalization:

1. Use representative training data.
2. Separate training and testing data.
3. Avoid overfitting.
4. Validate the model.
5. Update the model when fraud patterns change.

Example:

If a transaction has:

Unusual Location = Yes

Large Amount = Yes

New Device = Yes

the graphical model may calculate a high probability of fraud and flag the transaction.

Conclusion:

Graphical model learning captures relationships among fraud-related variables, while good generalization ensures that these learned patterns work on new transactions, improving fraud detection accuracy and reliability.